ABDILLAH ABHIPRAYA ABRAR

SOFTWARE DEVELOPER | APPLICATION SECURITY | TECHNICAL PROJECTS

+62 838-2913-7923 | abdillahabhi12@gmail.com | Yogyakarta, Indonesia

GitHub | Portfolio

PROFESSIONAL SUMMARY

Informatics Engineering undergraduate at UPN "Veteran" Yogyakarta with hands-on experience building and documenting software systems across web development, application security, real-time interfaces, and geospatial data processing. Technical work includes full-stack application development, QR-based authentication, ECDSA signature verification, backend/API integration, OCR prototyping, and WebGIS workflows. Brings a product-oriented approach to system analysis, software testing, UI/UX design, and technical problem-solving.

TECHNICAL SKILLS

Programming Languages: Python, C++, Java, PHP, JavaScript, HTML5, CSS3

Frontend & Frameworks: React 18, Vite, Tailwind CSS, Bootstrap, Framer Motion

Backend & Integration: Node.js, Express, FastAPI, REST APIs, Socket.IO / WebSockets, Android ADB

Databases & Data: PostgreSQL, JSON-based persistence, geospatial vector data, coordinate reference systems

Application Security: QR authentication, ECDSA P-256, ASN.1 DER signatures, token validation, origin validation, challenge expiry

Testing & Engineering: Backend test suites, Android unit testing, API integration, device testing, Git, GitHub, CI workflows

Design & 3D: Figma, Blender 3D, SketchUp, Adobe Photoshop, After Effects, Unity 3D

SELECTED TECHNICAL PROJECTS

IPGuard - Secure Digital Asset Protection Platform

2026 - Present

React | FastAPI | PostgreSQL | Android Integration | ECDSA P-256

- Developed full-stack components for a secure digital vault, connecting the React interface with FastAPI services and PostgreSQL persistence.
- Integrated Android QR sign-in with origin and payload validation, token-format checks, challenge-expiration controls, and ECDSA P-256 signature verification using ASN.1 DER.
- Verified integration with 65 passing backend tests and 37 passing Android unit tests; device enrollment and approved authentication sessions were confirmed.

MLBB Draft Studio - Real-Time Draft & OBS Overlay System

2026 - Present

React 18 | Vite | Tailwind CSS v4 | Express | Socket.IO | OBS Browser Source

- Built a real-time Mobile Legends draft-pick and scoreboard application with a control panel and browser-source overlays for OBS Studio.
- Designed server-authoritative state synchronization and JSON persistence; validated the application data pipeline containing 133 heroes.
- Prototyped hybrid OCR and image-recognition processing to extract game-state information from captured gameplay frames.

LahanAman - Landslide Susceptibility WebGIS

2026

Geospatial Processing | Vector QA | EPSG:32749 / EPSG:4326 | Weighted Scoring

- Performed reproducible auditing, processing, and integration of geospatial parameter layers for a landslide-susceptibility WebGIS.
- Validated six primary vector layers containing 963 features and reproduced a 917-feature susceptibility map using weighted scoring.

Local Government Budgeting System - Smart Validation & Happy Budgeting

2026

Requirements Engineering | Rule-Based Validation | Budgeting Workflows | UI/UX

- Defined requirements and workflows for a deterministic validation agent that checks local-government budget entries against structured rules; the validation approach is rule-based rather than ML-driven.
- Outlined guided budgeting and parameter-based recommendation flows, mapped user roles and functional requirements, and prepared system documentation aligned with ISO/IEC/IEEE 29148.

Image Steganography Inspection Tool

2026

Image Analysis | Steganography | Payload Inspection | In Development

- Developing the concept for an image-upload workflow that inspects steganography-encoded images and presents extractable embedded content to the user.
- Focuses on combining image input, payload extraction, and clear result presentation; implementation and validation are ongoing.

SmartHeight - Sensor-Based Measurement & Positioning Prototype 2026

Ultrasonic Sensors | Actuator Concept | Blender 3D | Mechanical Design | Design / Prototype

- Designed a measurement-device concept using dual SR04 ultrasonic sensors and an actuator to support object positioning and alignment.
- Explored multidirectional positioning (up/down, left/right, and front/back) and mechanical visualization of the prototype in Blender.

Personal Developer Portfolio 2025 - Present

Web Development | Cloudflare Pages | GitHub

- Designed and deployed a personal portfolio website to present software projects, technical interests, and visual work in a unified professional profile.

RESEARCH EXPERIENCE

Research Assistant - Financial Management Application Research May 2026 - Present

UPN "Veteran" Yogyakarta

- Contribute to research and development activities through system requirements analysis and software-architecture evaluation for a financial management application.
- Translate functional needs into Figma wireframes and UI/UX prototypes, supporting clear user flows and consistent system interactions.

LEADERSHIP EXPERIENCE

Deputy Head - Media & Information Department (Medinfo) Sep 2025 - Present

Badan Keluarga Mahasiswa Kebumen (KBMK)

- Coordinate digital-publication activities and editorial planning across Instagram, TikTok, and the organization's website; oversee multimedia deliverables and team workflows.

PDD Coordinator - Kebumen Campus Festival (KCF) 2026 Oct 2025 - Jan 2026

KCF 2026 Organizing Committee, Kebumen

- Led a nine-member creative and documentation team, coordinating photography, videography, graphic design, and copywriting across the event lifecycle.

PDD Coordinator - Serasehan Jurusan Informatika Aug 2026 - Present

Informatics Student Association, UPN "Veteran" Yogyakarta

- Lead documentation and visual-publication planning, coordinating photography, videography, and social-media content.

AWARDS, EDUCATION & LANGUAGES

Finalist & Best Video Award - Find IT! UGM 2026 May 2026

KMTETI, Faculty of Engineering, Universitas Gadjah Mada

Bachelor's Program in Informatics Engineering (In Progress) 2024 - Present

UPN "Veteran" Yogyakarta

Languages: Indonesian (Native) | English (Professional Working Proficiency)